

OLDS AGDM, AB, Canada

WMO: 712580

Lat: 51.7586N

Lon: 114.0847W

Elev: 1046

StdP: 89.37

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.5	-25.4	-31.8	0.2	-28.3	-28.6	0.3	-25.2	11.0	-5.7	9.8	-5.7	3.5	120	0.711

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	27.7	16.7	25.7	15.9	24.0	15.1	18.6	24.8	17.4	23.4	16.2	22.0	4.1	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.2	13.1	22.1	15.0	12.1	20.6	13.8	11.2	19.2	57.4	24.8	53.1	23.7	49.3	22.0	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.8	9.3	8.2	DB	-32.4	31.2	2.9	2.1	-34.5	32.7	-36.2	33.9	-37.8	35.1	-39.9	36.7	(4)
(5)			WB	-32.5	21.2	2.9	1.6	-34.6	22.4	-36.3	23.3	-37.9	24.2	-40.0	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.6	-7.3	-8.4	-3.1	3.4	9.5	13.2	16.1	15.0	10.6	4.5	-3.1	-8.2	(6)	
	DBStd	10.72	8.49	7.85	7.64	5.11	4.24	2.98	2.77	3.20	4.64	5.30	7.18	7.80	(7)	
	HDD10.0	2917	537	516	407	202	61	6	1	3	47	182	394	563	(8)	
	HDD18.3	5403	795	749	665	447	274	154	80	110	233	431	644	821	(9)	
	CDD10.0	577	0	0	1	5	45	103	189	158	66	11	1	0	(10)	
	CDD18.3	21	0	0	0	0	1	2	11	7	2	0	0	0	(11)	
	CDH23.3	519	0	0	0	2	31	45	193	168	78	3	0	0	(12)	
	CDH26.7	101	0	0	0	0	2	4	38	38	18	0	0	0	(13)	
(14) Wind	WSAvg	4.2	4.5	4.1	4.3	4.5	4.5	4.2	3.8	3.6	3.9	4.2	4.3	4.3	(14)	
(15) Precipitation	PrecAvg	499	21	15	22	27	64	97	81	70	46	21	19	21	(15)	
	PrecMax	629	41	39	45	98	127	180	213	114	104	43	39	48	(16)	
	PrecMin	314	2	1	3	4	14	20	28	16	5	5	6	5	(17)	
	PrecStd	74	11	8	11	19	27	38	37	29	25	11	10	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	8.7	14.9	21.9	26.3	26.9	29.8	30.2	29.2	23.0	15.6	9.1	(19)	
		MCWB	3.3	3.3	7.0	9.9	12.5	15.8	18.3	16.1	14.5	10.7	7.5	3.0	(20)	
	2%	DB	6.3	5.7	10.6	18.4	23.7	24.6	27.6	27.3	25.6	18.7	11.2	6.1	(21)	
		MCWB	2.2	1.7	4.6	8.6	11.7	15.0	18.5	16.2	13.9	9.7	5.2	2.2	(22)	
	5%	DB	5.0	3.5	8.0	15.0	21.2	22.5	25.5	25.0	22.6	15.5	8.0	4.1	(23)	
		MCWB	1.6	0.3	3.3	6.7	10.8	14.2	17.5	16.1	12.8	8.2	3.1	0.7	(24)	
	10%	DB	3.4	1.4	5.8	12.1	18.6	20.5	23.5	22.9	19.5	12.6	5.6	2.2	(25)	
		MCWB	0.4	-1.1	2.0	5.3	9.7	13.0	16.5	15.3	11.6	6.7	1.8	-0.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	3.9	7.6	10.5	13.7	18.3	21.0	19.2	15.7	12.2	7.9	3.9	(27)	
		MCDB	7.8	7.7	14.1	20.7	23.6	24.3	26.2	25.2	25.5	21.4	15.1	7.7	(28)	
	2%	WB	2.6	1.9	5.1	8.9	12.3	16.3	19.3	18.0	14.5	10.1	5.5	2.3	(29)	
		MCDB	5.9	5.3	9.5	17.7	21.7	22.2	25.6	24.1	23.7	17.8	10.7	6.0	(30)	
	5%	WB	1.7	0.5	3.5	7.2	11.3	14.9	18.1	16.7	13.3	8.5	3.7	0.9	(31)	
		MCDB	4.8	3.3	7.6	14.1	19.7	20.9	24.3	23.1	21.1	14.6	7.5	3.9	(32)	
	10%	WB	0.5	-0.9	2.1	5.6	10.2	13.7	17.0	15.6	12.0	7.0	2.0	-0.6	(33)	
		MCDB	3.4	1.2	5.5	11.3	17.4	19.3	22.5	21.8	19.1	12.2	5.0	2.1	(34)	
(35) Mean Daily Temperature Range	MDBR		9.7	9.8	9.9	11.4	12.6	11.4	12.6	13.0	12.7	11.7	9.8	9.1	(35)	
	5% DB	MCDBR	10.7	11.7	12.6	17.3	17.6	15.4	16.1	17.6	18.7	16.3	13.0	10.6	(36)	
		MCWBR	7.7	8.4	7.9	9.1	8.0	7.5	8.6	8.5	8.6	8.8	8.1	7.7	(37)	
	5% WB	MCDBR	10.5	11.2	12.2	16.1	15.3	13.5	15.0	15.4	17.0	15.7	12.7	10.4	(38)	
		MCWBR	7.7	8.2	8.0	8.8	7.3	7.2	8.6	8.3	8.1	8.6	8.2	7.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.253	0.263	0.282	0.330	0.338	0.343	0.342	0.355	0.308	0.277	0.249	0.238	(40)	
	taud		2.378	2.438	2.465	2.346	2.385	2.401	2.411	2.378	2.474	2.548	2.493	2.369	(41)	
	Ebn at Noon		783	883	928	909	915	910	905	872	885	857	795	743	(42)	
	Edh at Noon		58	73	87	111	112	112	109	107	86	65	51	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.07	3.41	4.63	5.51	5.85	6.31	5.19	3.65	2.23	1.19	0.85	(44)	
	RadStd		0.07	0.15	0.24	0.36	0.40	0.42	0.37	0.36	0.41	0.24	0.09	0.07	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page